MILITARY SPECIFICATION

BEEF GRAVY, POWDER
(FOR ORAL LIQUID FEEDING)

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers beef gravy powder for oral liquid feeding. This product is one component of a meal that, when consumed with other components in three meals or supplementary nourishments, provides a nutritionally adequate, orally consumed diet.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

JJJ-S-30	- Salad Oil, Vegetable

MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-P-44151	- Packaging and Meal Assembly of Powder (For Oral Liquid Feeding)

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Standards for Grades of Tomato Paste

(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Dairy Plants Surveyed and Approved for USDA Grading Practice

(Application for copies should be addressed to Dairy Standardization Section, Dairy Division, Room 2750-So., Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Production

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, P.O. Box 2109, Novato, CA 94947.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP) and the National Formulary (NF)

(Application for copies should be addressed to the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Beef, raw, ground. The beef shall be from steers, heifers, or cows and shall be derived from any of the following primal and subprimal cuts: shoulder clods, chuck rolls, ribs, ribeyes, trimmed full loins, trimmed short loins, strip loins, sirloins, top sirloin butts, rounds, top rounds, knuckles, and bottom rounds. Recognizable cuts are those which, when compared to IMPS cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. All suitable lean meat shall be used. The beef shall be in the fresh chilled state and shall be in excellent condition; i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class and cut of meat, and typical of meat that has been properly stored and handled. Cut surfaces shall show not more than slight darkening or discoloration due to dehydration, aging, and/or microbial activity. The fat shall show no more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, the meat shall show no evidence of freezing, defrosting, or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless, trimmed meat shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the same day as boning and trimming shall be maintained in the temperature range of 28° to 50°F (inclusive).

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b. Holding in the fresh chilled state not more than 4 days is permitted provided that the temperature of the beef is maintained in the temperature range of 28° to 40°F (inclusive).

c. Holding in the frozen state for not more than 60 days is permitted provided that the beef is:

-placed in freezer within 4 hours after certification,
-frozen to 0°F or lower within 72 hours and stored at 0°F or lower,
-protected from freezer deterioration and damage,
-stored in containers that are adequate to maintain product excellence, and
-maintained without defrosting until time of processing into the finished product.

3.2.2 Bromelain. The bromelain shall have a light tan to cream color, be readily soluble in water, and have a minimum activity of 800 Casein Digestion Units (CDUs) per milligram under assay conditions (see 4.5.1). It shall have a low odor typical of bromelain and be equivalent to the Enzeco Bromelain Concentrate manufactured by Enzyme Development Corporation.

3.2.3 Margarine. The margarine shall conform to the Federal Food, Drug, and Cosmetic Act Standards of Identity 21 CFR 166.40 and 166.110. The margarine shall have a sweet pleasing taste and aroma, free from rancidity or other objectionable flavors and odors. It shall have a medium yellow color and a smooth body and texture free of crumbliness, graininess, grittiness, oiliness, or droplets of free moisture. The margarine shall be salted. The margarine shall be in excellent condition, free from mold, slime, or discoloration. The margarine shall be manufactured in a plant listed in the Dairy Plants Surveyed and Approved for USDA Grading Service.

3.2.4 Tomato paste. Tomato paste shall be U.S. Grade A conforming to the U.S. Standards for Grades of Tomato Paste. The solids content shall be from 24 to 26 percent.

3.2.5 Oil, vegetable. Vegetable oil shall comply with the requirements of JJJ-S-30.

3.2.6 Onion powder. Onion powder shall comply with the fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

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3.2.7 Soup and gravy base, beef flavor. The beef-flavored base shall be a properly granulated mixture consisting of hydrolyzed vegetable protein and beef extracts or a beef extract alone, salt, sugar, soluble celery seasoning, MSG, onion powder, autolyzed yeast, hydrogenated vegetable shortening, caramel coloring, spices, disodium inosinate, and disodium guanylate. The base shall contain by weight not more than 46.0 percent nor less than 38.0 percent salt. It shall contain not more than 2.0 percent moisture, not more than 4.0 percent fat, not more than 1.5 percent soluble celery seasoning, and not more than 1.5 percent caramel color.

3.2.8 Burgundy flavor. Burgundy shall be a concentrated natural burgundy flavor. Concentration shall be such that 1 cup of concentrate would yield 8 cups of burgundy if fully reconstituted.

3.2.9 Caramel color. Caramel coloring shall be dry, readily soluble brown powder from sucrose or glucose. The dry caramel shall comply with the requirements of the Food Chemicals Codex.

3.2.10 Flour. Flour shall be white (bleached), clean, free from mold, and free from lumps that do not break under light finger pressure.

3.2.11 Starch, freeze-thaw stable. The starch shall be modified waxy maize type used in frozen processed foods. The cooked starch slurry shall have a bland flavor. It shall be white, odorless, and finely pulverized.

3.2.12 Methylcellulose. Methylcellulose shall be a white, free-flowing powder with no off odors or flavors. It shall have a viscosity of 4,000 centipoises in a 2 percent aqueous solution as measured at a temperature of 68°F by a Ubbelohde tube for viscosity measurements. The viscosity shall be certified by a certificate of conformance provided by the manufacturer. Methylcellulose shall comply with the requirements of the Food Chemicals Codex.

3.2.13 Water. Water used for formulation and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and processing.

3.3.1 Beef processing.

a. The boneless meat shall be ground at least once through a plate having holes no larger 0.75 inch in diameter (or it may be otherwise reduced in size provided that the texture and appearance of the product after final grinding is typical of ground beef prepared by grinding only). Final grinding shall be through a plate having holes 1/16 inch in diameter.

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b. Fat content shall be not less than 5.0 percent and not more than 15.0 percent (see 4.5b).

3.3.1.1 Handling of meat. Processing, formulation, and cooking shall be continuous after the grinding of the meat.

3.3.2 Product formulation. The ground beef and other ingredients shall be formulated as follows:

Ingredients	Percent by weight
Water	45.8
Beef, raw, ground through a 1/16-inch plate	25.0
Bromelain	0.1
Margarine	7.7
Tomato paste	2.3
Oil, vegetable	1.9
Onion powder	1.7
Soup and gravy base, beef flavor	1.2
Burgundy flavor	0.3
Caramel color	0.2
Flour	1.0
Starch, freeze-thaw stable	1.9
Water, cold, for starch slurry	1.9
Methylcellulose	0.4
Water, hot (176°-194°F), for methylcellulose	4.3
Water, cold (40°-50°F), for methylcellulose	4.3

3.3.2.1 Amount of water. The proportion of water in the formulation is specified to enable the activation and inactivation of the bromelain, to cook the product, to make the starch and flour slurry, and to disperse the methylcellulose. These amounts may be increased or decreased to facilitate cooking, making dispersions, pumping, or changing the viscosity before freeze dehydration.

3.3.3 Cooking. The water shall be heated to 130°F (±5°F). The beef and bromelain shall be added and the mixture shall be simmered at 130°F (±5°F) while being stirred for 20 minutes. The mixture shall then be cooked at 200°F for 15 minutes to inactivate the bromelain. (Note: After inactivating the bromelain at 200°F for 15 minutes, the Azocoll protease assay (see 4.5.2) may be conducted to determine if the bromelain has been inactivated. If the enzyme has not been inactivated, the product may be reheated once for an additional 15-minute period at 200°F and the Azocoll protease assay repeated to ensure inactivation of the bromelain. However, conducting the assay

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procedure at this stage does not substitute for the requirement of 3.4.2e.) The margarine, tomato paste, oil, onion powder, soup and gravy base, burgundy flavor, and caramel color shall be added and mixed well. The mixture shall be heated until the margarine is completely melted. A slurry shall be made by mixing together the flour, starch, and a small amount of cold water. This slurry shall be added to the beef mixture, and the mixture shall be heated to 180°F while being stirred. The product shall be held at that temperature for 10 minutes. The heat shall then be turned off, and the mixture cooled rapidly to 120°F. Methylcellulose shall be prepared by mixing with hot water (176°-194°F) until all particles are wetted and a smooth paste is obtained. The cold water (40°-50°F) shall be added to this methylcellulose paste, and stirring shall continue until a smooth slurry is obtained. The methylcellulose slurry shall be combined with the other ingredients and stirred until the mixture is smooth. The mixture shall be frozen to 0°F or below within 12 hours after cooking is completed.

3.3.4 Freezing. The product shall be portioned onto freeze dryer trays immediately after cooling and shall be frozen to 0°F or below within 12 hours after cooking is completed. The product shall be freeze dehydrated within 48 hours after being frozen.

3.3.5 Freeze dehydration. The product shall be freeze dehydrated at an absolute pressure not to exceed 1.5 millimeters of mercury. The product temperature, as indicated by a suitable instrument, shall be no more than 150°F, and the platen temperature shall be 155°F. After dehydration has been completed, the pressure shall be equalized to atmospheric level with nitrogen, and the product shall be pulverized. During the interim, the product shall be adequately protected from oxygen and moisture by being held either under a nitrogen atmosphere of 2.0 percent oxygen or less or under a vacuum of at least 27.0 inches of mercury for the entire period. If a vacuum is used, it shall be broken with nitrogen. The nitrogen shall comply with U.S. Pharmacopeia grade as specified in the United States Pharmacopeia.

3.3.6 Pulverization. After freeze dehydration, the product shall be pulverized through a U.S. Standard No. 10 sieve (0.0787 inch).

3.3.7 Packaging. After pulverization, the product shall be packaged without delay in pouches as specified in MIL-P-44151. In no case shall more than 16 hours elapse from the time the freeze dehydration chamber is opened to the time the product is completely packaged, and at no time shall more than 3 hours elapse from the time the product is pulverized to the time it is completely packaged.

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3.4 Finished product requirements.

3.4.1 Dehydrated, pulverized product. The dehydrated product shall comply with the following requirements:

a. The product shall contain no foreign or extraneous material.

b. The product shall be free from caking and hard lumps.

c. The product shall leave no particles remaining when passed through a U.S. Standard No. 10 sieve.

d. The moisture content shall not exceed 2.0 percent (see 4.5.3).

e. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample (see 4.5.4).

f. The Escherichia coli count shall be less than 3 per gram (no positives in the standard 3 tube MPN technique) in four of five samples and not greater than 20 per gram in any sample (see 4.5.4).

g. The sample average net weight of the lot shall be not less than 45.0 grams.

h. The individual pouches shall contain not more than 50.0 grams nor less than 40.0 grams.

3.4.2 Rehydrated product. Rehydration of the dehydrated powder for oral liquid feeding shall be accomplished by mixing each pouch (45 grams) with 1 cup (8 ounces) of hot water in a high-speed food blender for 30 seconds. For a pureed diet, only 1/2 cup (4 ounces) of liquid should be used. The rehydrated product shall comply with the following requirements:

a. After rehydration for oral liquid feeding, no more than 0.5 gram of product shall remain when the product has passed through a U.S. Standard No. 10 sieve.

b. The rehydrated product shall have the flavor and odor characteristic of beef gravy.

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c. The rehydrated product shall have a light brown color.

d. For oral liquid feeding, the rehydrated product shall have a consistency that allows it to be easily sipped through a straw. For a pureed diet, the viscosity shall be semisolid so as to be edible with a spoon.

e. Representative samples of the dehydrated powder (100 mg each) or of the rehydrated liquid product (0.1 mL each) shall show no enzymatic activity in the Azocoll protease assay (see 4.5.2).

3.4.3 Palatability. The rehydrated product shall be equal to or better than the approved preproduction sample in palatability and overall appearance (see 4.3, 6.2, and 6.3).

3.5 Plant qualifications. The meat component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.1.1 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 6.2), it shall be inspected in accordance with the quality assurance provisions of MIL-P-44151 and this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall comply with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected to comply with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or other applicable purchase documents.

4.4.1.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

4.4.1.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the AMS agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot if, in his or her opinion, that portion may be out of compliance with any requirements. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef; the sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or presence of any off-condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

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TABLE I. Sampling and acceptance criteria for boned and trimmed beef

Lot size (pounds)	Sample size (No. of sample units)	Defect categories			
		Major		Minor	
		AC	RE	AC	RE
500 or less	13	1	2	3	4
501 to 1200	20	2	3	5	6
1201 to 3200	32	3	4	7	8
3201 to 10,000	50	5	6	10	11
10,001 to 35,000	80	7	8	14	15
Reinspection lots of 10,001 to 35,000	125	10	11	21	22

TABLE II. Boned and trimmed beef defects

Category		Defect 1/ 2/
Major	Minor	
101		Presence of lymph gland measuring 1.0 inch or more in any dimension.
	201	Presence of lymph gland measuring 0.5 inch or more but less than 1.0 inch in any dimension.
102		Presence of blood clot measuring 1.0 inch or more in any dimension.

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TABLE II. Boned and trimmed beef defects - Continued

Category		Defect 1/ 2/
Major	Minor	
	202	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension.
103		Presence of bruise measuring 1.0 inch or more in any dimension.
	203	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
104		Presence of bone measuring 0.3 inch or more in any dimension.
105		Presence of cartilage measuring 0.5 inch or more in any dimension.
106		Presence of backstrap measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	204	Presence of backstrap measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
107		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more.
	205	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 0.5 square inch or more but less than 2.0 square inches.

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TABLE II. Boned and trimmed beef defects - Continued

Category		Defect 1/ 2/
Major	Minor	
	206	Presence of heavy connective tissue on lower edge of short plate or flank.
	207	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	208	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3 square inches or more.
108		Presence of knuckle cover.
109		Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of kidney, pizzle eye, prepubic tendon, hanging tender, or thymus gland measuring less than 1.0 inch in one dimension and 0.2 inch in a second dimension (when measured at right angles to each other).
	210	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.
	211	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	212	Presence of dehydrated surface measuring 1.0 square inch or more.

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TABLE II. Boned and trimmed beef defects - Continued

Category		Defect 1/ 2/
Major	Minor	
	213	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more.
	214	Presence of exposed major blood vessel measuring 1.0 inch or more in any dimension.
	215	Presence of cod, udder, kidney, or pelvic fat.

1/ Acceptability of product with respect to the presence of foreign material (for example, glass, dirt, insect parts, hair, wood, metal) shall be made by a USDA employee. The presence of any product not in excellent condition shall be cause for rejection of the lot.

2/ Evidence of freezing or defrosting or presence of any off-condition (including but not restricted to being off-odor, slightly sticky, gassy, rancid, or sour, or showing discoloration) shall be cause for rejection of the involved lot.

4.4.1.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance shall be evident by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such supplier-furnished valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically, as necessary, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.4.1.4 Filled and sealed pouch examination. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44151.

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4.4.2 In-process examination. In-process examination shall be performed to determine conformance to preparation and processing requirements in 3.3. Any nonconformance revealed by actual examination or by review of record of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

4.4.3 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Results shall be reported to the nearest 0.1 gram. Any individual net weight of less than 40.0 grams or more than 50.0 grams shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3, and the acceptable quality level (AQL) shall be 2.5, expressed in terms of percent defective. In addition, the lot shall be rejected if the sample average net weight is less than 45.0 grams.

4.4.4 Finished product inspection. The finished product shall be examined for the defects listed in table III. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3, and the AQL shall be 1.5, expressed in terms of defects per hundred units.

TABLE III. Product characteristics 1/ 2/

Category	Defect
Major	Applicable to dehydrated product
111	Product is not free from caking and hard lumps.
112	Presence of particles remain on a U.S. Standard No. 10 sieve. 3/
	Applicable to rehydrated product 4/
113	More than 0.5 gram of product does not pass through a U.S. Standard No. 40 sieve. 5/
114	The rehydrated product does not have flavor or odor normal for beef gravy.

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TABLE III. Product characteristics - Continued

Category	Defect
Major	Applicable to dehydrated product
115	The rehydrated product does not have a light brown color.
116	The viscosity is not acceptable for appropriate use of the product.

1/ The presence of any foreign material (e.g., glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale, etc.), or foreign color shall be cause for the rejection of the whole lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.2) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by a USDA inspection agent.)

3/ The powder shall be poured onto a U.S. Standard No. 10 sieve, and compressed air shall be blown through the product. The presence of visible particles remaining on the sieve surface shall be considered a defect.

4/ Rehydration of the dehydrated powder for oral liquid feeding shall be accomplished by mixing each pouch (45 grams) with 1 cup (8 ounces) of hot water in a high-speed blender for 30 seconds. For a pureed diet, use 1/2 cup (4 ounces) of liquid.

5/ Particle size determination shall be made on the sample(s) after rehydration for oral liquid feeding (use a different sample for organoleptic examination). The rehydrated product shall be poured onto a U.S. Standard No. 40 screen. The product shall be washed for 1 1/2 minutes with water at a temperature of 96° to 100°F that is running at a flow of 90 to 100 mL/sec through 1/2-inch tubing (interior dimension). The sieve shall be allowed to drain for two minutes, then excess water may be wiped from bottom and sides of the sieve. The remaining solids shall be blown from the sieve with compressed air onto a tared weighing paper (see 3.4.2).

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4.4.5 Packaging. Packaging shall be in accordance with the applicable requirements of MIL-P-44151.

4.5 Product testing. The following tests shall be administered to this product:

a. The bromelain powder used in preparing this product shall be tested as specified in 4.5.1 before using it in preparing this product. The lot size shall be expressed in grams. The sample unit shall be 50.0 mg. Four samples shall be selected at random from each lot. If test results are less than 800 CDUs per milligram, the bromelain is considered inactive and the enzyme is unacceptable. Any test failure shall be cause for rejection of the bromelain component.

b. The raw ground beef shall be tested for fat content in accordance with the crude fat or ether extract method in the Meat and Meat Products chapter of the Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC). Four samples shall be selected at random from each lot. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirements of 3.3.1 shall be cause for rejection of the lot.

c. The finished dehydrated powder shall be tested for enzymatic activity as specified in 4.5.2, for moisture content as specified in 4.5.3, and for microbiological requirements as specified in 4.5.4.

4.5.1 Bromelain assay. (To determine the activity of the bromelain ingredient, see 3.2.2.)

a. The following components shall be prepared for tests:

(1) Enzyme diluent - Dissolve 5.3 g cysteine hydrochloride monohydrate and 2.2 g disodium EDTA dihydrate in 900 mL of distilled water and adjust to a pH of 4.5 by adding 0.1 N HCL or 1 N NaOH. Add sufficient distilled water to produce 1000 mL. The enzyme diluent shall be prepared fresh before each assay.

(2) Casein substrate - Dissolve 0.6 g Hammarsten casein in 80 mL 0.05 M Na_2HPO_4 and adjust to a pH of 7.0 by adding 0.1 N HCL or 0.1 N NaOH. Dilute this mixture to 100 mL by adding the necessary amount of distilled water. The casein substrate shall be made fresh before each assay.

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(3) Protein precipitant: Trichloroacetic acid (TCA) - Dissolve 18 g CCl_3COOH, 29.9 g CH_3COONa, and 19.8 g CH_3COOH in sufficient distilled water to produce 1000 mL of TCA. (WARNING: Avoid contact of TCA with skin.)

(4) Diluted bromelain enzyme solution - Accurately weigh 50.0 mg of bromelain powder and dissolve it in 100 mL of the enzyme diluent. Dilute this solution with distilled water 25 times (100 mL will yield 2500 mL by volume). This enzyme solution shall be prepared fresh before each assay. The dilution amount (in this example, 25) is the value for B in equation of 4.5.1c.

b. The test procedure shall be as follows:

(1) Pipette 1.0 mL of diluted bromelain enzyme solution into a test tube and place in a water bath at 37°C for about 3 minutes.

(2) To the enzyme solution, add 5 mL of casein substrate prewarmed to 37°C. Stir and incubate at 37°C for exactly 10 minutes.

(3) At the end of incubation, terminate the reaction by adding 5 mL of TCA. Shake vigorously and let stand for 30 minutes to allow complete precipitation of the protein material.

(4) Remove the resulting precipitates by filtration. Set spectrophotometer at 275 nm and read the absorbance of the filtrate; this is Et in the calculations.

(5) For a blank test, mix the enzyme solution with TCA before adding the casein substrate. Read the absorbance of the filtrate at 275 nm; this is Eb in the calculations.

(6) Read the absorbance (Es), also at 275 nm, of tyrosine in 0.1 N HCL containing 50 micrograms tyrosine per milliliter.

c. Calculations shall be according to the following equation:

$$\text{Casein Digestion Units CDU/mg} = \frac{Et - Eb}{Es} \times 50 \times \frac{11}{10} \times \frac{100 \times B}{\text{Enzyme wt. (50 mg)}}$$

NOTE: "B" in this equation is the dilution factor applied to the 50 mg/100mL bromelain solution (in this example, B = 25). The dilution may need to be adjusted to result in 40 to 50 units in the dilution solution.

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4.5.2 Azocoll protease assay. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The sample size shall be the number of sample units indicated by inspection level S-3. If test results are less than +0.01 absorbance units per 100 mg sample, the enzyme is considered inactive and the product is acceptable. Any test failure shall be cause for rejection of the finished dehydrated powder. (To determine if bromelain has been inactivated after cooking and/or processing, see 3.3.3 and 3.4.2e.)

a. The following components shall be prepared for the test:

(1) 1 M citrate stock - Mix together 210 g citric acid monohydrate, 750 mL deionized water, and 55 g NaOH and adjust to a pH of 4.6 by adding more NaOH if necessary. Dilute this mixture to 1000 mL by adding the necessary amount of deionized water. Add 100 mg NaAzide (a preservative) and mix well.

(2) 0.05 M citrate buffer - Combine 25 mL of the 1 M citrate stock solution with approximately 470 mL deionized water and adjust to a pH of 4.8 by adding 1 N NaOH or 1 N HCL if necessary. Dilute this mixture to 500 mL by adding deionized water.

(3) Activator (100 mM Cleland's Reagent) - Add sufficient deionized water to 155 mg of Cleland's Reagent (dithiothreitol) to make a 10 mL solution. This solution shall be prepared fresh before each assay.

b. The test procedure shall be as follows:

(1) Set up a series of test tubes as listed in table IV.

(2) Mix ingredients in each test tube well, then place in a 50°C shaking bath for 10 minutes.

(3) After 10 minutes, start reaction at one-minute intervals by adding 100 mg of the sample (beef gravy powder), mixing well after each addition and returning the test tubes to the water bath. Add the sample only to tubes 1, 2, and 3. Tube 4 is a blank.

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TABLE IV. Test tube contents for Azocoll Protease assay

Tube No.	Identity	0.05 M citrate buffer (mL)	Activator (mL)	Azocoll 1/ (mg)
1	sample #1	4.75	0.25	20
2	sample #2	4.75	0.25	20
3	meat blank	4.75	0.25	-
4	blank	4.75	0.25	20

1/ Hide Powder Azure may be used in place of Azocoll.

(4) Wait 15 minutes. Pull tubes from the water bath at one-minute intervals and mix. Stop the reaction by filtering by gravity for two minutes through Whatman No. 54 (9 cm) paper into clean test tubes. Filtrate should be clear for accurate spectrophotometer readings.

(5) Set spectrophotometer at 520 nm (595 nm if using Hide Powder Azure) and take readings.

(6) Subtract the total of the absorbances of both blanks from the absorbance of the sample. If the remainder averages +0.01 or higher, the enzyme has not been inactivated. If the remainder averages less than +0.01, the enzyme is considered inactivated.

c. Refer to table V for examples of calculations for this assay.

TABLE V. Sample test calculations

Tube No.	Identity	Abs 520 (example 1) 1/	Abs 520 (example 2) 2/
1	sample #1	0.522	0.165
2	sample #2	0.502	0.167
3	meat blank	0.091	0.081
4	blank	0.081	0.091

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TABLE V. Sample test calculations - Continued

1/ Calculations (example 1):

0.522 - (0.091 + 0.081) = +0.35
0.502 - (0.091 + 0.081) = +0.33
Average = +0.34

Conclusion: enzyme not inactivated

2/ Calculations (example 2):

0.165 - (0.081 + 0.091) = -0.007
0.167 - (0.081 + 0.091) = -0.005
Average = -0.006

Conclusion: enzyme inactivated

4.5.3 Moisture content testing. The product shall be tested for moisture content in accordance with the Moisture in Dried Fruits Method (vacuum drying) in the Fruit and Fruit Product Chapter in the Official Methods of Analysis of the AOAC with the exception that the product shall be dried for 16 hours. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4.1 shall be considered a defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed laminated pouch. The inspection level shall be S-2 and the AQL shall be 1.5, expressed in terms of percent defective.

4.5.4 Microbiological testing. Five filled and sealed pouches of the product shall be selected at random from the lot regardless of lot size. The contents of each sample pouch shall be microbiologically tested, as applicable, in accordance with the Microbiological Methods Chapter of the Official Methods of Analysis of the AOAC using the section on Examination of Frozen, Chilled, Precooked, or Prepared Foods-Official First Action. The diluent shall be added to each sample of dry product and allowed to stand for 15 minutes before the blending of that sample. Continue as directed by AOAC. Any test failure shall be cause for rejection of the lot. Any of the following results shall be considered a test failure:

a. Any of the five samples has an aerobic plate count greater than 150,000 per gram.

b. Two of the five samples have an aerobic plate count greater than 75,000 per gram.

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c. Any of the five samples has a confirmed E. coli count of greater than 20 per gram.

d. Two of the five samples have a confirmed E. coli count equal to or greater than 3 per gram.

5. PACKAGING

5.1 Packaging. Packaging shall comply with the applicable requirements of MIL-P-44151.

5.2 Labeling and marking.

5.2.1 Pouches. The pouches shall be labeled as follows and examined for compliance in accordance with MIL-P-44151.

a. BEEF GRAVY, POWDER (FOR ORAL LIQUID FEEDING)

b. MENU NO. 3 (DINNER)

c. List of ingredients in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

d. Mixing instructions, as follows:
"Empty contents into a high-speed food blender.
Add 1 (8 ounces) cup hot water.
Blend for 30 seconds.
(For a pureed diet, use 1/2 cup (4 ounces) of liquid.)"

e. The net weight, as follows: "45 grams net."

f. Contractor's name and address.

g. USDA inspection marking (as required).

h. Date markings in accordance with MIL-L-1497.

i. Code.

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6. NOTES

6.1 Intended use. This product is intended to be used by patients requiring an oral liquid or pureed diet (e.g., those with maxillofacial injuries, wired jaws, oral surgery, dental disease, or swallowing difficulties).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Provisions for preproduction samples (see 3.4.3 and 6.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Custodian:

Army - GL

Preparing activity:

Army - GL

Project No. 8940-A473

Review activities:

Army - MD, TS
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44169(GL)	Beef Gravy, Powder (For Oral Liquid Feeding)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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